understanding high-level causal structures. CRL techniques could enhance interpretability by identifying causal relationships between neurons or layers (Bengio et al., 2019; Ke et al., 2021), potentially revealing model reasoning. Its focus on interventions and counterfactuals (Pearl & Mackenzie, 2018; Peters et al., 2017) could inspire new methods for probing model internals (Goyal et al., 2020; Besserve et al., 2019). CRL's emphasis on learning invariant representations (Peters et al., 2015; von Kügelgen et al., 2019) could guide the search for robust features, while its approach to transfer learning (Rojas-Carulla et al., 2018; Magliacane et al., 2018) could inform studies into model generalization.

iii.) **Trojan Detection**: Detecting deceptive alignment models is a key motivation for inspecting model internals, as – by definition – deception is not salient from observing behavior alone (Casper et al., 2024). However, quantifying progress is challenging due to the lack of evidence for deception as an emergent capability in current models (Steinhardt, 2023), apart from *sycophancy* (Sharma et al., 2023; Denison et al., 2024) and theoretical evidence for *deceptive inflation* behavior (Lang et al., 2024). Detecting trojans (or backdoors) (Hubinger et al., 2024) implanted via data poisoning could be a proxy goal and proof-of-concept. These trojans simulate *outer misalignment* (where the model's behavior is misaligned with the specified reward function or objectives due to poorly defined or incorrect reward signals) rather than *inner misalignment* such as deceptive alignment (where the model appears aligned with the specified objectives but internally pursues different, misaligned goals). Moreover, activating a trojan typically results in an immediate change of behavior, while deception can be subtle, gradual, and, at first, entirely internal. Nevertheless, trojan detection can still provide a practical testbed for benchmarking interpretability methods (Maloyan et al., 2024).

iv.) **Adversarial Robustness**: There is a duality between interpretability and adversarial robustness (Elhage et al., 2022b; Räuker et al., 2023; Bereska, 2024). More interpretable models tend to be more robust against adversarial attacks (Jyoti et al., 2022), and vice versa, adversarially trained models are often more interpretable (Engstrom et al., 2019). For instance, techniques like input gradient regularization have been shown to simultaneously improve the interpretability of saliency maps and enhance adversarial robustness (Ross & Doshi-Velez, 2017; Du et al., 2021). Furthermore, interpretability tools can help create more sophisticated adversaries (Carter et al., 2019; Casper et al., 2021), improving our understanding of model internals. Viewing adversarial examples as inherent neural network *features* (Ilyas et al., 2019) rather than bugs also hints at alien features beyond human perception. Connecting mechanistic interpretability to adversarial robustness thus promises ways to gain theoretical insight, measure progress (Casper, 2023), design inherently more robust architectures (Fort & Lakshminarayanan, 2024), and create interpretability-guided approaches for identifying (and mitigating) adversarial vulnerabilities (García-Carrasco et al., 2024).

More details on the interplay between interpretability, robustness, modularity, continual learning, network compression, and the human visual system can be found in the review by Räuker et al. (2023).

Corroborate or Refute Core Assumptions. Features are the fundamental units defining neural representations and enabling mechanistic interpretability's bottom-up approach (Chan, 2023), but defining them involves assumptions requiring scrutiny, as they shape interpretations and research directions. Questioning hypotheses by seeking additional evidence or counter-examples is crucial.

The *linear representation* hypothesis treats activation directions as features (Park et al., 2023a; Nanda et al., 2023b; Elhage et al., 2022b), but the emergence and necessity of linearity is unclear – is it architectural bias or inherent? Stronger theory justifying linearity's necessity or counter-examples like autoencoders on uncorrelated data without intermediate linear layers (Elhage et al., 2022b) are needed. An alternative lens views features as polytopes from piecewise linear activations (Black et al., 2022), questioning if direction simplification suffices or added polytope complexity aids interpretability.

The *superposition* hypothesis suggests that *polysemantic* neurons arise from the network compressing and representing many features within its limited set of neurons (Elhage et al., 2022b), but polysemanticity can also occur incidentally due to redundancy (Lecomte et al., 2023; Marshall & Kirchner, 2024; McGrath et al., 2023). Understanding superposition's role could inform mitigating polysemanticity via regularization

(Lecomte et al., 2023). Superposition also raises open questions like operationalizing *computation in superposition* (Vaintrob et al., 2024; Hänni et al., 2024), *attention head superposition* (Elhage et al., 2022b; Jermyn et al., 2023; Lieberum et al., 2023; Gould et al., 2023), representing feature clusters (Elhage et al., 2022b), connections to adversarial robustness (Elhage et al., 2022b; García-Carrasco et al., 2024; Bloom & Bailey, 2023), anti-correlated feature organization (Elhage et al., 2022b), and architectural effects (Nanda, 2023a).

8.2 Setting Standards

Prioritizing Robustness over Capability Advancement. As the mechanistic interpretability community expands, it is essential to maintain the norm of not advancing AI capabilities while simultaneously establishing metrics necessary for the field's progress (Räuker et al., 2023). Researchers should prioritize developing comprehensive tools for analyzing the worst-case performance of AI systems, ensuring robustness and reliability in critical applications. This includes focusing on adversarial tasks, such as backdoor detection and removal (Lamparth & Reuel, 2023; Hubinger et al., 2024; Wu et al., 2022a), and evaluating the accuracy of explanations in producing adversarial examples (Goldowsky-Dill et al., 2023).

Establishing Metrics, Benchmarks, and Algorithmic Testbeds. A central challenge in mechanistic interpretability is the lack of rigorous evaluation methods. Relying solely on intuition can lead to conflating hypotheses with conclusions, resulting in cherry-picking and optimizing for best-case rather than average or worst-case performance (Rudin, 2019; Miller, 2019; Räuker et al., 2023; Casper, 2023). Current ad hoc practices and proxy measures (Doshi-Velez & Kim, 2017) risk over-optimization (Goodhart's law – *When a measure becomes a target, it ceases to be a good measure*). Distinguishing correlation from causation is crucial, as interpretability illusions demonstrate that visualizations may be meaningless without causal linking (Bolukbasi et al., 2021; Friedman et al., 2023a; Olah et al., 2017).

To advance the field, rigorous evaluation methods are needed. These should include: *(i)* assessing out-of-distribution inputs, as most current methods are only valid for specific examples or datasets (Räuker et al., 2023; Ilyas et al., 2019; Mu & Andreas, 2020; Casper et al., 2023c; Burns et al., 2023); *(ii)* controlling systems through edits, such as implanting or removing trojans (Mazeika et al., 2022) or targeted editing (Ghorbani & Zou, 2020; Dai et al., 2022; Meng et al., 2022a;b; Bau et al., 2018; Hase et al., 2023); *(iii)* replacing components with simpler reverse-engineered alternatives (Lindner et al., 2023); and *(iv)* comprehensive evaluation through replacing components with hypothesized circuits (Quirke et al., 2024).

Algorithmic testbeds are essential for evaluating faithfulness (Jacovi & Goldberg, 2020; Hanna et al., 2024) and falsifiability (Leavitt & Morcos, 2020). Tools like Tracr (Lindner et al., 2023) can provide ground truth labels for benchmarking search methods (Goldowsky-Dill et al., 2023), while toy models studying superposition in computation (Vaintrob et al., 2024) and transformers on algorithmic tasks can quantify sparsity and test intrinsic methods. Recently, Thurnherr & Scheurer (2024); Gupta et al. (2024) introduced datasets of transformer weights with known circuits for evaluating mechanistic interpretability techniques.

8.3 Scaling Techniques

Broader and Deeper Coverage of Complex Models and Behaviors. A primary goal in scaling mechanistic interpretability is pushing the Pareto frontier between model and task complexity and the coverage of interpretability techniques (Chan, 2023). While efforts have focused on larger models, it is equally crucial to scale to more complex tasks and provide comprehensive explanations essential for provable safety (Tegmark & Omohundro, 2023; Dalrymple et al., 2024; Gross et al., 2024) and enumerative safety (Cunningham et al., 2024; Elhage et al., 2022b) by ensuring models won't engage in dangerous behaviors like deception. Future work should aim for thorough *reverse engineering* (Quirke & Barez, 2023), integrating proven modules into larger networks (Nanda et al., 2023a), and capturing sequences encoded in hidden states beyond immediate predictions (Pal et al., 2023). Deepening analysis complexity is also key, validating the realism of toy models (Elhage et al., 2022b) and extending techniques like path patching (Goldowsky-Dill et al., 2023; Liu et al., 2023a) to larger language models. The field must move beyond small transformers on algorithmic tasks (Nanda et al., 2023a) and limited scenarios (Friedman et al., 2023a) to tackle more complex, realistic cases.

Towards Universality. As mechanistic interpretability matures, the field must transition from isolated empirical findings to developing overarching theories and universal reasoning primitives beyond specific circuits, aiming for a comprehensive understanding of AI capabilities. While collecting empirical data remains valuable (Nanda, 2023f), establishing motifs, empirical laws, and theories capturing universal model behavior aspects is crucial. This may involve finding more circuits/features (Nanda, 2022a;c), exploring circuits as a lens for memorization/generalization (Hanna et al., 2023), identifying primitive general reasoning skills (Feng & Steinhardt, 2023), generalizing specific findings to model-agnostic phenomena (Merullo et al., 2023), and investigating emergent model generality across neural network classes (Ivanitskiy et al., 2023). Identifying universal reasoning patterns and unifying theories is key to advancing interpretability.

Automation. Implementing automated methods is crucial for scaling interpretability of real-world state-of-the-art models across size, task complexity, behavior coverage, and analysis time (Hobbhahn, 2022). Manual circuit identification is labor-intensive (Lieberum et al., 2023), so automated techniques like circuit discovery and sparse autoencoders can enhance the process (Foote et al., 2023; Nanda, 2023b). Future work should automatically create varying datasets for understanding circuit functionality (Conmy et al., 2023), develop automated hypothesis search (Goldowsky-Dill et al., 2023), and investigate attention head/MLP interplay (Monea et al., 2023). Scaling sparse autoencoders to extract high-quality features automatically for frontier models is critical (Bricken et al., 2023). Still, it requires caution regarding potential downsides like AI iteration outpacing training (RicG, 2023) and loss of human interpretability from tool complexity (Doshi-Velez & Kim, 2017).

8.4 Expanding Scope

Interpretability Across Training. While mechanistic interpretability of final trained models is a prerequisite, the field should also advance interpretability before and during training by studying learning dynamics (Nanda, 2022b; Elhage et al., 2022b; Hubinger, 2022). This includes tracking neuron development (Liu et al., 2021), analyzing neuron set changes with scale (Michaud et al., 2023), and investigating emergent computations (Quirke & Barez, 2023). Studying phase transitions could yield safety insights for *reward hacking* risks (Olsson et al., 2022).

Multi-Level Analysis. Complementing the predominant bottom-up methods (Hanna et al., 2023), mechanistic interpretability should explore top-down and hybrid approaches, a promising yet neglected avenue. The top-down analysis offers a tractable way to study large models and guide microscopic research with macroscopic observations (Variengien & Winsor, 2023). Its computational efficiency could enable extensive "comparative anatomy" of diverse models, revealing high-level motifs underlying abilities. These motifs could serve as analysis units for understanding internal modifications from techniques like instruction fine-tuning (Ouyang et al., 2022) and reinforcement learning from human feedback (Christiano et al., 2017; Bai et al., 2022).

New Frontiers: Vision, Multimodal, and Reinforcement Learning Models. While some mechanistic interpretability has explored convolutional neural networks for vision (Cammarata et al., 2021; 2020), vision-language models (Palit et al., 2023; Salin et al., 2022; Hilton et al., 2020), and multimodal neurons (Goh et al., 2021), little work has focused on vision transformers (Palit et al., 2023; Aflalo et al., 2022; Vilas et al., 2023; Pan et al., 2024). Future efforts could identify mechanisms within vision-language models, mirroring progress in unimodal language models (Nanda et al., 2023a; Wang et al., 2023).

Reinforcement learning (RL) is also a crucial frontier given its role in advanced AI training via techniques like reinforcement learning from human feedback (RLHF) (Christiano et al., 2017; Bai et al., 2022), despite potentially posing significant safety risks (Bereska & Gavves, 2023; Casper et al., 2023a). Interpretability of RL should investigate reward/goal representations (Mini et al., 2023; Colognese & Jozdien, 2023; Colognese, 2023; Bloom & Colognese, 2023; Bloom & Bailey, 2023), study circuitry changes from alignment algorithms (Prakash et al., 2024; Jain et al., 2023; Lee et al., 2024; Jain et al., 2024), and explore emergent subgoals